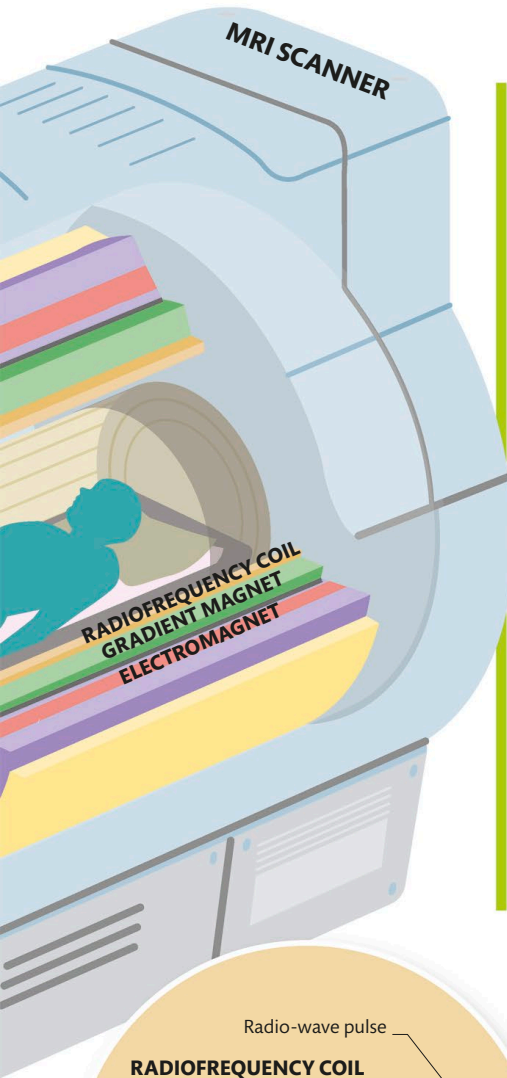
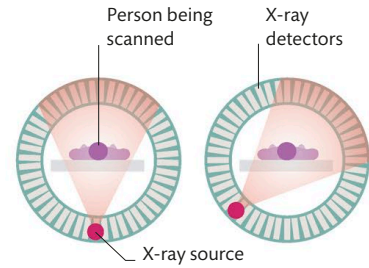




CT scans

Computer tomography (CT), or computerized axial tomography (CAT), takes a series of X-ray images through the brain from different angles. A computer then compares the images to create a single cross section of the brain. CT scans are quicker than MRI and are best for detecting strokes, skull fractures, and brain hemorrhages.



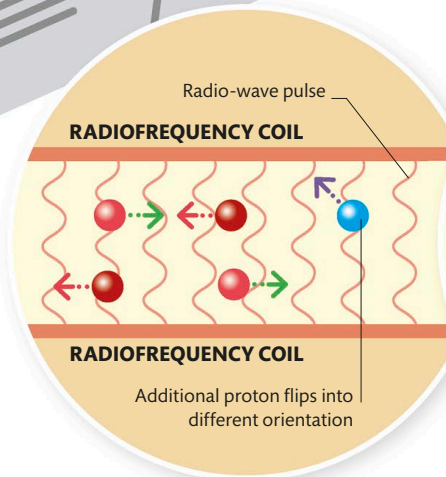
Rotating X-ray

The X-ray source shines through the brain, arcing around the patient to vary the angle of each image.

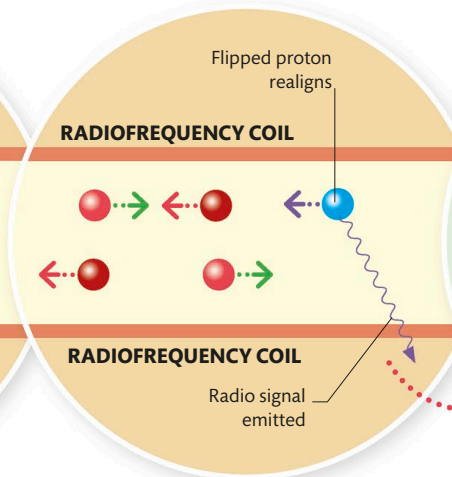
OTHER TYPES OF SCANNING TECHNOLOGY

Imaging certain brain features require particular scanning techniques, which may also be used if MRI or CT are dangerous or unsuitable.

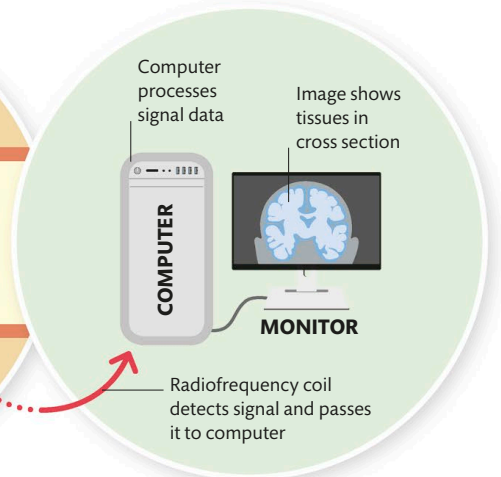
TYPE OF SCAN	TECHNOLOGY AND USES
PET (positron emission tomography)	Used in order to image the blood flow through the brain and highlight active regions. PET scans track the location of radioactive tracers injected into the blood.
DOI (diffuse optical imaging)	An array of newer techniques that works by detecting how bright light or infrared rays penetrate into the brain. DOI provides a way of monitoring blood flow and brain activity.
Cranial ultrasound	A safe imaging technique that is based on the way ultrasonic waves bounce off structures in the brain. Cranial ultrasound is mostly used on infants. It is used less often in adults because the images lack detail.



- 3 A pulse of radio waves**
With the magnetic field on, the MRI machine's radiofrequency coil sends a pulse of radio waves through the brain. This input of extra energy makes the spare protons flip out of alignment.



- 4 Radio signal emitted**
Once the pulse is switched off, the unaligned protons flip back into alignment with the magnetic field. This causes them to release energy as a radio signal, which is detected by the machine.



- 5 Receiver creates image**
All the signal data is then processed by computer to create two-dimensional "slices" of the brain. Protons in different body tissues produce different signals, so scans can show the tissues distinctly and in great detail.